Experiment 11 - Experiment to understand pandas library use cases in Data Science.

Aim:

To explore and execute various applications of the pandas library within data science.

Program:

```
import numpy as np
import pandas as pd

list = [[1, 'Smith', 50000], [2, 'Jones', 60000]]
df = pd.DataFrame(list)
df

df.columns = ['Empd', 'Name', 'Salary']
df
df.info()

df = pd.read_csv("/content/drive/MyDrive/50_Startups.csv")
df.info()
df.head()
df.tail()

import numpy as np
import pandas as pd

df = pd.read_csv("/content/drive/MyDrive/employee.csv")
df.head()
df.tail()
df.info()

df.salary
type(df.salary)
df.salary.mean()
df.salary.median()
df.salary.mode()
df.salary.var()
df.salary.std()
df.describe()
```

```
df.describe(include='all')
empCol = df.columns
empCol

emparray = df.values
emparray
employee_DF = pd.DataFrame(emparray, columns=empCol)
```

Output:

	0	1	2
0	1	Smith	50000
1	2	Jones	60000

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     object
 4   Profit           50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116983.80	45173.06	California	14681.40

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

	emp id	name	salary
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  7 non-null      int64
 1   name    7 non-null      object
 2   salary  7 non-null      int64
dtypes: int64(2), object(1)
memory usage: 300.0+ bytes
```

	salary
0	5000
1	6000
2	7000
3	5000
4	8000
5	3000
6	6000

dtype: int64

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy:
bool | None=None, fastpath: bool | lib.NoDefault=lib.no_default) -> None
```

One-dimensional ndarray with axis labels (including time series).

Labels need not be unique but must be a hashable type. The object
supports both integer- and label-based indexing and provides a host of
methods for performing operations involving the index. Statistical
methods from ndarray have been overridden to automatically exclude

```
np.float64(5714.285714285715)
```

```
6000.0
```

	salary
0	5000
1	6000

dtype: int64

```
2571428.5714285714
```

```
1603.5674514745463
```

	emp id	salary
count	7.000000	7.000000
mean	4.000000	5714.285714
std	2.160247	1603.567451
min	1.000000	3000.000000
25%	2.500000	5000.000000
50%	4.000000	6000.000000
75%	5.500000	6500.000000
max	7.000000	8000.000000

	emp id	name	salary
count	7.000000	7	7.000000
unique	NaN	6	NaN
top	NaN	SRI HARSHAVARDHANAN R	NaN
freq	NaN	2	NaN
mean	4.000000	NaN	5714.285714
std	2.160247	NaN	1603.567451
min	1.000000	NaN	3000.000000
25%	2.500000	NaN	5000.000000
50%	4.000000	NaN	6000.000000
75%	5.500000	NaN	6500.000000
max	7.000000	NaN	8000.000000

```
Index(['emp id', 'name ', 'salary'], dtype='object')
```

```
array([[1, 'SREE VARSSINI K S', 5000],
       [2, 'SREEMATHI B', 6000],
       [3, 'SREYA G', 7000],
       [4, 'SREYASKARI MULLAPUDI', 5000],
       [5, 'SRI AKASH U G', 8000],
       [6, 'SRI HARSHAVARDHANAN R', 3000],
       [7, 'SRI HARSHAVARDHANAN R', 6000]], dtype=object)
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

Result:

Thus, an experiment to understand pandas library use cases in Data science was conducted.